# CELESTA

## SPACE MAGAZINE

VOL
01





# Editor's Note

Somewhere between "I liked the Moon landing documentary" and "I can recite the Drake Equation at parties," there exists a large and delightful population of people who are simply interested in space. This magazine is for you, whether you're a card-carrying astrophysicist who winces at our simplifications, or someone who just found out Venus spins the wrong way and needs to tell everyone immediately. We've built this issue the way you'd build a good telescope: something for the casual stargazer, something sturdy enough for the serious ones, and absolutely nothing that requires you to already own a lab coat. Expect news, myths cheerfully dismantled, a defensible amount of nerding out about black holes, and the occasional joke that would have made Carl Sagan sigh and Douglas Adams nod approvingly. Read it in order or don't — the universe has no beginning we can point to either, so we won't hold it against you.

— Aradhya Haldikar | President

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# Meet the Editorial Board

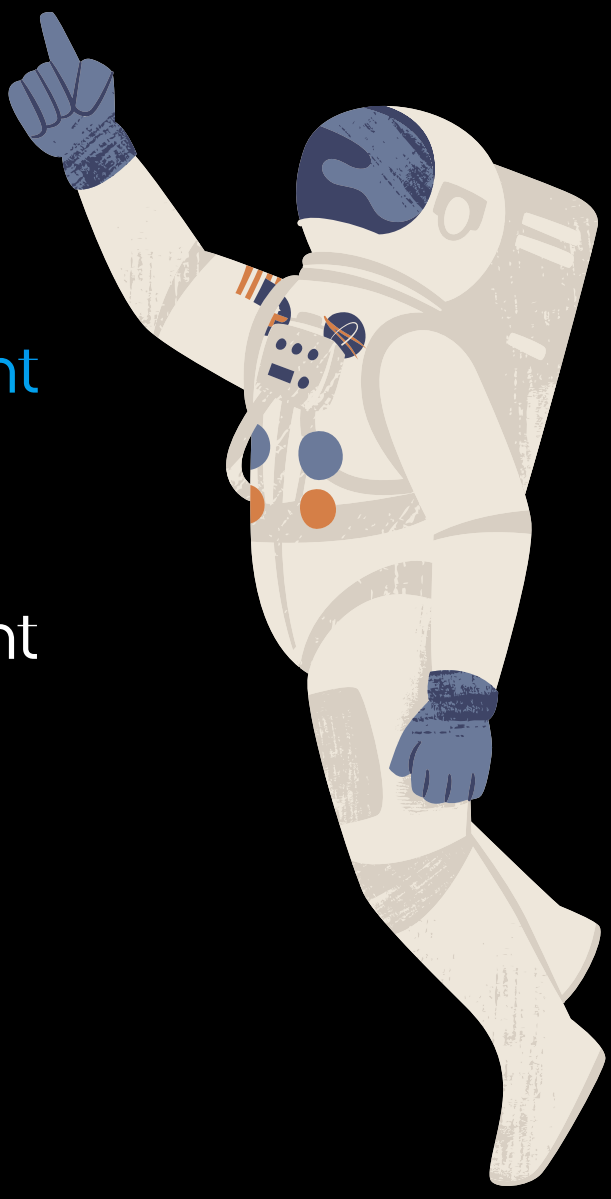
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# Theme of the Month : New Frontiers

This issue's theme is new frontiers, chosen mostly because 2026 refuses to slow down: private companies are launching people into orbit with the regularity of a commuter train, and entire nations are edging toward their first crewed missions. Every article here, one way or another, is about someone doing something for the first time and hoping gravity doesn't notice.

# Quotes by Famous Space Personalities

"The Earth is the cradle of humanity, but mankind cannot stay in the cradle forever." — Konstantin Tsiolkovsky

"Somewhere, something incredible is waiting to be known." — Carl Sagan

"We choose to go to the Moon in this decade and do the other things, not because they are easy, but because they are hard." — John F. Kennedy

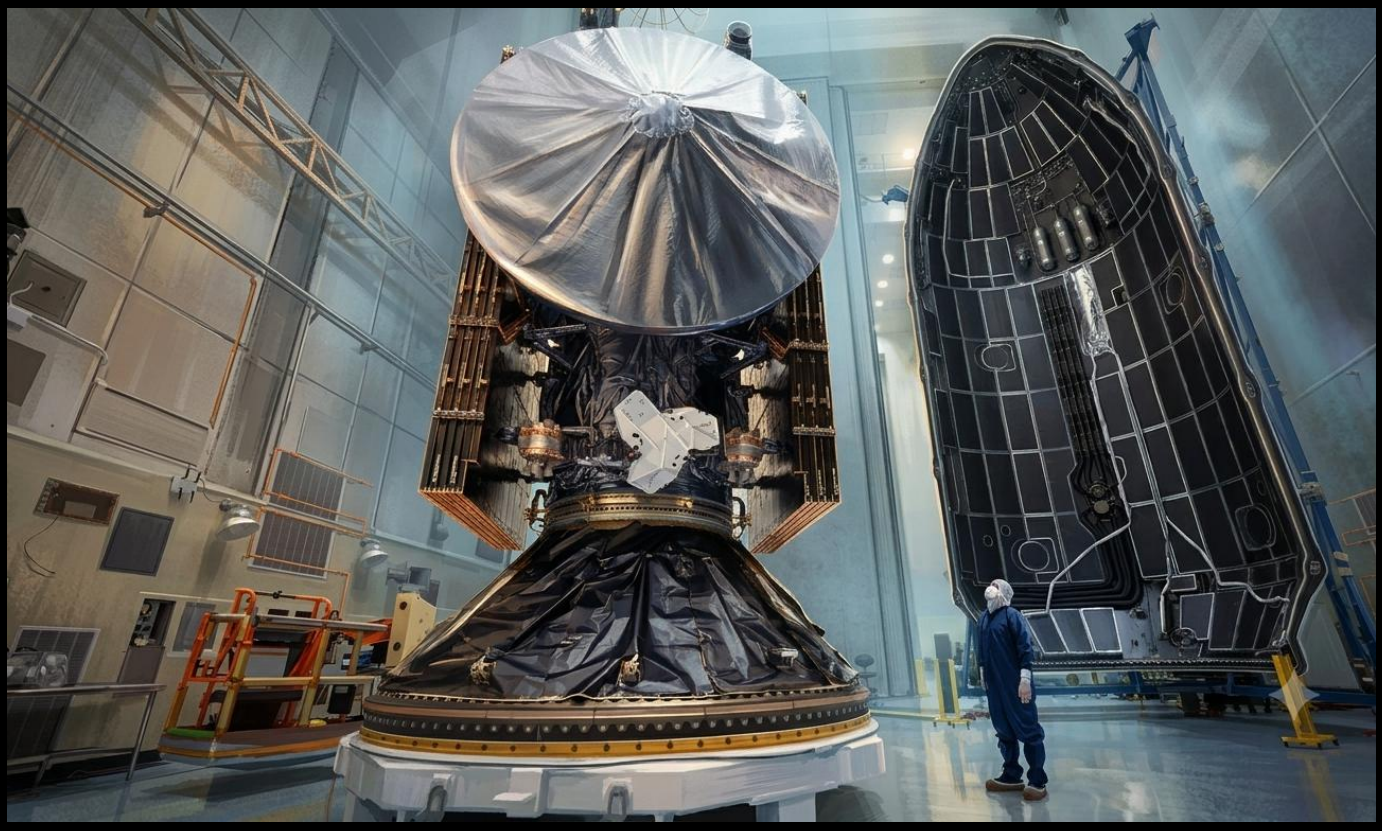
# Launch Pad



Space, it turns out, does not take August off. A total solar eclipse swept across Greenland, Iceland, and Spain on August 12 — mainland Europe's first proper look at totality in decades, and by all accounts the sort of thing that makes grown scientists forget their instruments for a full ninety seconds.

Meanwhile, SpaceX has been treating orbit like a subscription service, clearing ninety Falcon 9 launches for the year by early August, mostly hauling up Starlink satellites with the occasional classified payload for the U.S. National Reconnaissance Office tucked in for variety. Starship, not to be left out, flew a mostly successful test in late July — "mostly" doing a great deal of quiet, diplomatic work in that sentence — with another booster catch expected before the year wraps up.

Elsewhere, a 30-year NASA veteran, astronaut Mike Fincke, announced his retirement after flying on four different spaceflights — a career so varied it reads like a spacecraft loyalty card. And two newer rockets, Firefly's Alpha Block 2 and Rocket Lab's Neutron, continue their slow, deliberate creep toward an actual launch date, in the time-honored tradition of "any day now."



# Mission Spotlight

## Europa Clipper: Hunting for a Hidden Ocean

NASA built its largest planetary spacecraft ever, launched it in October 2024, and pointed it at a moon that isn't even the biggest one in its own neighborhood. Europa doesn't need size to be interesting, though — under its ice sits what may be an ocean containing more water than every ocean on Earth combined, which is either wonderful news for future dolphins or slightly humbling for the rest of us.

Clipper isn't landing — it's circling, close and often, once it arrives at Jupiter around 2030, using its instruments to measure the ice shell's thickness, taste the moon's thin atmosphere, and go hunting for organic molecules. Recent research suggests that ice shell might be tougher than expected, a plot twist that only sharpens the central question: is there something down there worth finding, or is Europa just very good at keeping secrets?

Nine science instruments, the largest solar arrays ever flown on a NASA planetary mission, and roughly fifty flybys planned once it settles in — Clipper is, in short, showing up to Jupiter extremely well prepared.

# Cosmic Figure

## Vera Rubin — The Astronomer Who Found the Universe's Missing Mass

Vera Rubin spent her career watching stars go around in circles, which sounds like a mundane hobby until you realize she was the one who noticed they were going around far too fast. In the 1970s, working with Kent Ford, Rubin measured how quickly stars orbited within spiral galaxies and found that the outer stars were moving at speeds that should have flung them into deep space — if only the visible matter in the galaxy were all there was holding them in.



It wasn't. Something invisible was doing the heavy lifting, gravitationally speaking, and Rubin's careful, unglamorous measurements are largely why we now take dark matter seriously — a substance that makes up roughly 27% of the universe and has never once bothered to show itself directly.

The Nobel committee, in one of astronomy's most quietly infuriating oversights, never got around to Rubin. She did receive the National Medal of Science and spent much of her later life championing women in the field. Fittingly, her name now belongs to an observatory in Chile designed to photograph the entire visible sky every few nights — the sort of relentless, patient watching that would have made her very proud, and possibly a little smug.

# Cosmic Debate

## Should Humans Colonize Mars?

### Arguments For

- **Insurance policy for the species.** Keeping all of humanity on one planet is, statistically, not what a risk manager would recommend.
- **The science alone justifies it.** A permanent human presence would allow research far beyond what a rover trundling along at the pace of a caffeinated tortoise can manage.
- **Necessity breeds invention.** Life support, closed-loop farming, radiation shielding — the demands of Mars tend to produce technology that quietly improves life back on the home planet.

### Arguments Against

- **We haven't finished this one yet.** Critics point out that the resources required to colonize Mars could do a great deal of good addressing problems — climate change, poverty, disease — on the planet we're already standing on.
- **Mars is, frankly, trying to kill you.** Toxic thin air, brutal radiation, temperatures that make Antarctica look tropical, and dust so fine it gets into everything, including — allegedly — the will to continue.
- **Nobody's quite sure who's in charge.** Governance, land rights, and legal responsibility on Mars remain unsettled questions, and history suggests "figure it out later" is not humanity's strongest legal strategy.

Take a side, argue it and post it on social media tagging us!
We might feature you in our next edition...

# Space Phenomenon

## Time Dilation: Why Clocks Don't Agree

Einstein's relativity comes with a genuinely unsettling little footnote: time does not tick at the same rate for everyone, everywhere. Move fast enough, or sit deep enough in a gravitational well, and your clock quietly falls out of step with everyone else's — not because it's broken, but because time itself is behaving differently for you.

This isn't a thought experiment confined to chalkboards. GPS satellites orbit fast enough, and far enough from Earth's gravity, that their onboard clocks drift out of sync with ground clocks by about 38 microseconds a day. Left uncorrected, your GPS would cheerfully route you into a lake within about twenty-four hours.

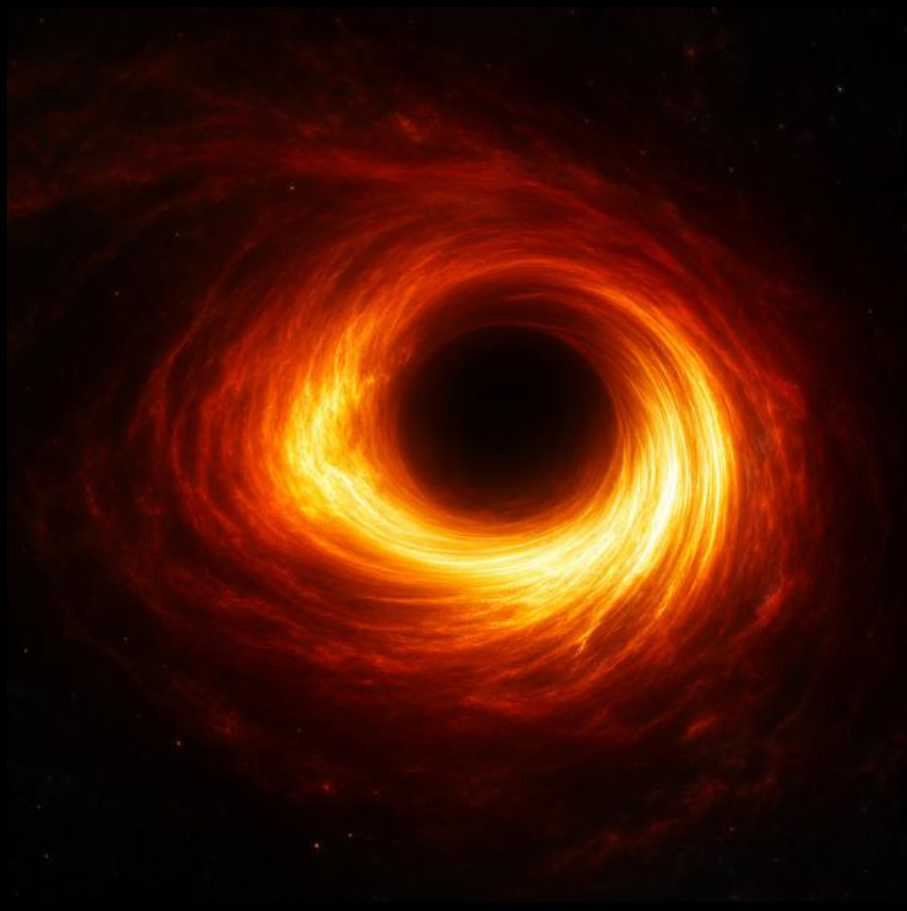
Astronauts on the International Space Station get their own small, strange souvenir from all this: thanks to their orbital speed, they age a few milliseconds less over a six-month mission than they would have on the ground — the world's most expensive way to very slightly outrun a birthday.

# Deep Space
# Black Holes, Explained

A black hole forms when an enormous amount of matter gets crushed into a very small space, creating gravity so overwhelming that nothing — not light, not radio waves, not a very persuasive argument — can escape once it crosses a boundary called the event horizon. Past that line, every possible direction, mathematically speaking, leads further in.

There are two main varieties. **Stellar-mass black holes** form when a star many times heavier than our Sun exhausts its fuel and collapses in on itself, often with a supernova as a parting shot. **Supermassive black holes**, millions to billions of times heavier still, sit quietly at the centers of most large galaxies — including our own, home to Sagittarius A*, a black hole doing its best impression of a very well-behaved houseguest.

Since black holes emit no light of their own, astronomers find them the hard way: by watching them bend light from objects behind them, by tracking stars orbiting something invisible, or, in the most dramatic case yet, by imaging the glowing ring of superheated gas swirling just outside the event horizon — as the Event Horizon Telescope did for the black holes at the hearts of galaxy M87 and our own Milky Way.



Black holes also do strange things to time: to an outside observer, a clock falling toward the event horizon appears to slow down and eventually freeze in place — general relativity's way of reminding us that even something as basic as "now" gets negotiable near enough gravity.

*Fun footnote: The event horizon isn't a physical wall — it's a mathematical line in the sand. And in theory, black holes eventually evaporate entirely through a slow process called Hawking radiation, on timescales that make "eventually" feel almost sarcastic.*

# ISRO CORNER



2026 has turned into one of ISRO's busiest years on record, headlined by preparations for the first uncrewed **Gaganyaan (G1)** mission — flying on the Human-Rated LVM3 rocket, with the humanoid robot Vyommitra along for the ride to test life-support systems ahead of India's first crewed spaceflight, targeted for 2027. Somewhere, a robot is quietly doing the most important test flight of its short existence and getting none of the credit.

Beyond Gaganyaan, ISRO is targeting roughly seven missions this financial year — its most ambitious calendar yet — spanning Earth observation satellites, commercial rideshare missions through NewSpace India Limited, and continued qualification testing for the Gaganyaan crew module, including water-recovery trials completed earlier this year.

Further out on the horizon: payload selection is underway for **Shukrayaan-1**, India's first dedicated mission to Venus, which will carry a synthetic aperture radar, infrared camera, and atmospheric spectrometers to study a planet whose atmosphere would crush and cook most things we care about. And quietly, in the background, planning continues for the **Bharatiya Antariksha Station** — India's own future space station, still very much a work in progress, in the best sense.

All of it builds on the momentum of Chandrayaan-3's 2023 south-pole Moon landing and the Aditya-L1 solar mission — proof that India's space program has stopped asking for a seat at the table and started building its own.

# Space Around the World

**NASA (USA)** is juggling Artemis-era lunar preparation with a packed ISS schedule — spacewalks this August alone to prep hardware for a new solar array — while keeping missions like Europa Clipper and the Parker Solar Probe running in the background, because apparently one to-do list wasn't enough.
**SpaceX (USA)** remains the commercial launch industry's most caffeinated member, running a Falcon 9 cadence of multiple launches a week, testing Starship, and ferrying crews to the ISS with the casual confidence of a company that has simply decided rockets are routine now.









**ESA (Europe)** had a front-row seat to August's total solar eclipse and continues its partnership on JUICE, currently coasting toward Jupiter's icy moons on a journey patient enough to make continental drift look rushed.

**CNSA (China)** continues expanding Tiangong station operations and pressing ahead on lunar exploration ambitions, as part of a long-term, unhurried plan to put its own astronauts on the Moon.

**Rocket Lab and Firefly Aerospace (USA)** are both racing new medium-lift rockets — Neutron and Alpha Block 2 respectively — toward their debut flights, hoping to chip away at SpaceX's near-monopoly on getting things into orbit.

# Cosmic Mythbusters

## Myth: "You'd Instantly Explode in the Vacuum of Space"

A wonderfully cinematic idea, and almost entirely wrong. Human skin is, against all odds, strong enough to hold your body together even as the pressure around it drops to zero — no dramatic pop, no scattering of confetti-you across the void.

What actually happens is quietly worse. Without pressure, gases dissolved in your blood start forming bubbles, not unlike a bad case of the bends. Any air still in your lungs will expand rapidly if you don't exhale first. And within about fifteen seconds, you'll lose consciousness as your blood — now boiling at body temperature, thanks to the sudden pressure drop — stops delivering oxygen to your brain. Rapid repressurization within a minute or two can, remarkably, still be survivable, as a few unlucky high-altitude accidents have demonstrated.

So: no explosion, no fireworks. Just a very fast, very quiet, and thoroughly unpleasant way for physics to make its point. Space, it turns out, doesn't need Hollywood's help.

# Instrument of the Month

## The Spectrograph: Reading Starlight Like a Fingerprint



Nobody has ever touched a star and lived to describe the experience, and yet astronomers will happily tell you exactly what one is made of. The trick is the spectrograph — an instrument that splits incoming light into its component wavelengths, the same way a prism turns sunlight into a rainbow, except with rather more riding on the outcome.

Pass starlight through a spectrograph and you get a spectrum crossed with dark or bright lines at very specific wavelengths — each chemical element's signature, as distinctive as a fingerprint and considerably harder to fake. Match the lines, and you know the star's composition, its temperature, and even how it's moving through space.

That last trick relies on the Doppler effect: spectral lines shifted toward red mean a star is moving away, shifted toward blue means it's headed our way. It's the same principle behind many of the earliest exoplanet discoveries — astronomers watching a star's spectrum wobble ever so slightly as an invisible planet tugs on it from the shadows, like a very patient, very cosmic game of hide and seek.

# Then vs Now

Four hundred years is not, cosmically speaking, very long. And yet in that time we went from squinting at Jupiter's moons through a glorified spyglass to reading the chemical ingredients of atmospheres on planets we will never see with the naked eye.

| | **Then — Galileo, 1609** | **Now — James Webb Space Telescope** |
|---|---|---|
| Aperture | ~4 cm lens | 6.5 meter primary mirror |
| Location | Handheld, ground-based | 1.5 million km from Earth, at L2 |
| Could see | Craters on the Moon, Jupiter's four largest moons, the phases of Venus | Galaxies over 13 billion light-years away, exoplanet atmospheres, stellar nurseries in infrared |
| Biggest challenge | Convincing people Earth wasn't the center of the universe | Detecting molecules in the atmospheres of worlds we can't even directly photograph |
| Data recorded | Hand-drawn sketches, occasionally squinted at twice for good measure | Multi-gigapixel digital images, shared with the world within hours |

Both efforts, four centuries apart, boiled down to the same basic act of stubbornness: pointing something at the sky and refusing to stop looking.

# Space Laws Simplified

**Who owns the Moon?** No one — and by treaty, no one ever will, however many flags get planted on it. The 1967 Outer Space Treaty explicitly forbids any nation from claiming sovereignty over the Moon or any other celestial body, flags notwithstanding.

**Can countries claim planets?** Also no. The same treaty bars national appropriation of any part of outer space "by claim of sovereignty, by means of use or occupation, or by any other means" — a clause that has aged remarkably well.

**Space mining laws** occupy a murkier middle ground. Several countries, including the U.S. and Luxembourg, have passed domestic laws letting companies own resources they extract from asteroids or the Moon — even though no nation can own the celestial body those resources came from. Whether this squares neatly with international treaty law is a debate that shows no sign of resolving itself soon.

**The Artemis Accords** are a set of non-binding principles, led by NASA and signed by dozens of countries, laying out shared norms for lunar exploration — including "safety zones" around lunar operations and support for extracting and using space resources.

**The Outer Space Treaty (1967)** remains the foundation of it all: it bans weapons of mass destruction in orbit, prohibits territorial claims, and holds nations responsible for their activities in space, including those of private companies flying under their flag.

**Space debris responsibility** falls under the 1972 Liability Convention, which makes a launching nation liable for damage its space objects — including any debris — cause, whether on Earth or in orbit.

**Satellite regulations**, meanwhile, are coordinated internationally through the ITU, which assigns frequencies and orbital slots so satellites don't end up talking over one another like guests at a badly organized dinner party.

**Weaponization of space** is partially addressed: the Outer Space Treaty bans nuclear weapons and other weapons of mass destruction in orbit or on celestial bodies, but stops short of banning all military activity in space — leaving plenty of room for ongoing argument about anti-satellite weapons and space-based defense systems.

# Did You Know?

Venus spins backwards compared to almost every other planet — so if you were standing on its surface (you would not survive this, but bear with the thought experiment), the Sun would rise in the west and set in the east.

A day on Venus, at 243 Earth days, is longer than its entire year, which lasts a comparatively brisk 225 Earth days — the planet has, in effect, given up trying to keep its own calendar straight.

One exoplanet, 55 Cancri e, is theorized to be partly composed of diamond — which is either the universe's idea of a joke or the most impractical engagement ring imaginable.

Neutron stars are so dense that a single teaspoon of their material would weigh roughly as much as a mountain — proof that the universe has a sense of proportion, and has chosen to ignore it entirely

The footprints astronauts left on the Moon will likely last for millions of years. No wind, no rain, no mischievous younger sibling — just quiet, permanent evidence that someone was there.

# Sky This Month

## The October 2026 Sky

This month offers a packed calendar for skywatchers. Based on bulletins from NASA and the Royal Observatory Greenwich, here is what to look for throughout the month.

**Draconid Meteor Shower: 8–9 October**
Active from 6 to 10 October, the Draconids reach peak activity on the night of 8–9 October. Originating from debris shed by Comet 21P/Giacobini–Zinner, this shower is best viewed during the early evening rather than pre-dawn hours, as its radiant constellation, Draco, sits highest immediately after twilight. Under clear, dark skies, observers can anticipate roughly 10 meteors per hour.

**Planetary Twilight: Mercury and Venus: 7–12 October**
On the evening of 7 October, Mercury and Venus form a low-lying pair above the western horizon just after sunset, offering the clearest views to watchers at lower latitudes and in the Southern Hemisphere. On 12 October, Mercury reaches its greatest eastern elongation, placing it at its maximum angular separation from the Sun and marking the month's premier window for spotting the elusive planet.

**Orionid Meteor Shower: 21–22 October**
The Orionids peak overnight on 21–22 October, driven by remnant dust from Comet 1P/Halley. The Royal Observatory Greenwich reported these meteors strike Earth's upper atmosphere at roughly 66 kilometres per second, generating fast streaks and occasional fireballs. While the shower yields around 15 meteors per hour under pristine skies, moonlight may wash out fainter streaks; observers will find the best conditions in the early morning hours following moonset.

**The Hunter's Moon: 26 October**
The full Hunter's Moon officially arrives at 04:12 UTC on 26 October. Historically marking the autumn period when Northern Hemisphere communities preserved meat ahead of winter, the rising Moon often exhibits a deep orange or reddish hue caused by light scattering through the denser lower atmosphere near the horizon.



# The India High School Exoplanet Data Challenge

In 2026, Celesta organized the **India High School Exoplanet Data Challenge**, a free, online machine learning hackathon designed for Indian high school students in Grades 9–12.

Hosted on Devpost, the challenge ran from **June 15 to July 31, 2026,** giving participants approximately one and a half months to work through an open-ended machine learning problem using real astronomical data.

Students could participate individually or in teams of up to three. Rather than relying on a simplified educational dataset, the challenge was built around data derived from the **NASA Exoplanet Archive's KOI Cumulative Table**, giving participants experience with the same kinds of imperfect, highdimensional data that make real-world scientific machine learning difficult.

The koi_score field was removed, and empty columns were deliberately retained, requiring participants to confront data cleaning and preparation as part of the challenge rather than treating the dataset as ready-made.

## What Participants Had to Do

The challenge was structured around the full machine-learning workflow rather than simply maximizing prediction accuracy.

Participants were expected to work through:

Data cleaning
↓
Exploratory Data Analysis
↓
Feature Engineering
↓
Model Development
↓
Evaluation
↓
Explainability
↓
Final Technical Write-up

The judging rubric reflected this structure:

| Category | Weight |
|---|---|
| Model Performance | 35% |
| EDA & Data Handling | 20% |
| Feature Engineering | 20% |
| Explainability | 15% |
| Write-up Quality | 10% |

This weighting encouraged participants to demonstrate not only whether a model worked, but also whether they understood the data, the features they used, the reasoning behind their model, and the limitations of their approach.

# Participation

The challenge closed with **484 participants overall**.

The organizers had initially set a target of approximately 400 participants, while the event began with fewer than 10 sign-ups. The final participation therefore substantially exceeded the original target.

The challenge received approximately **50 credible submissions**, giving the judging panel a substantial set of projects to evaluate.

# Featherless.ai

The event was supported by **Featherless.ai**, a serverless AI inference platform.

Featherless contributed directly to the competition's prize structure and provided participants with free credits for experimentation.

Every participant was eligible for **$25 in Featherless credits**, representing approximately **$12,000 in credits was made available across participants.**

Featherless also funded the **$300 in credits awarded to the first-place team**. The sponsorship therefore served two purposes: supporting the competition's prize structure while also giving high-school participants access to AI infrastructure that could otherwise be difficult to obtain.

# The Judging Panel

The competition was evaluated by a **10-member judging panel** bringing together experience from software engineering, artificial intelligence, research, astronomy, data science, and competitive STEM programs.

Judging was conducted offline using a custom Google Sheets-based evaluation tracker rather than relying on Devpost's native judging system.

# Final Results

| Award | **Participant** | **Project** |
|---|---|---|
| 1st Place | Avyaay Rathi | *Hunting Exoplanets with Machine Learning* |
| 2nd Place | Hritesh Srivastava | *Celestivine* |
| 3rd Place | Adhyansh Verma | *Kepler-AI* |
| Best Explain ability | Mukilan Senthilkumar | *Predicting KOI Disposition Without NASA's Vetting Flags* |
| Rising Star | Shaurya Ranjan | *ExoJury* |

# Recognition

The competition also recognized projects that demonstrated strengths beyond overall placement. The **Best Explainability** award highlighted work that placed particular emphasis on understanding and communicating how a machine-learning system reached its conclusions. The **Rising Star** award recognized a promising project with notable potential.

# Why the Challenge Mattered

The India High School Exoplanet Data Challenge combined three areas that are often separated in school-level STEM programs: astronomy, machine learning, and real scientific data.

Students were not simply asked to identify exoplanets through a predetermined classroom exercise. They had to work with a substantial astronomical dataset, make decisions about data preparation and features, build and evaluate models, explain their results, and present their work.

For high-school students, this created an opportunity to experience a simplified version of the workflow used in data-driven scientific research.

The challenge also demonstrated the scale that student-led scientific organizations can reach when education, technology, community-building, and external partnerships are brought together.

# CelestaGPT

CelestaGPT is an AI-based astronomy mentor developed as part of Celesta's educational initiatives. The tool is designed to help students explore astronomy and related STEM topics through conversational guidance.

Its intended areas of use include:

- Astronomy and astrophysics concepts
- Telescope and observational guidance
- Astrophotography
- Astronomy Olympiad preparation
- Advanced problem-solving
- General astronomy learning and exploration

CelestaGPT reflects a broader objective of the organization: using technology to reduce the barriers that can prevent students from accessing specialized astronomy knowledge.

# Celesta By The Numbers

**484**
Total hackathon participants

**10**
Judges from technology, engineering, AI, research and astronomy backgrounds

**$25**
Featherless credits made available to every participant

**~$12,000**
Approximate total value of participant Featherless credits

**$300**
Featherless-funded first-place credits

**88**
Discord community members as of September 2026

**3**
Core social/digital platforms used by Celesta

**3**
Maximum team size for the hackathon

# Student Corner

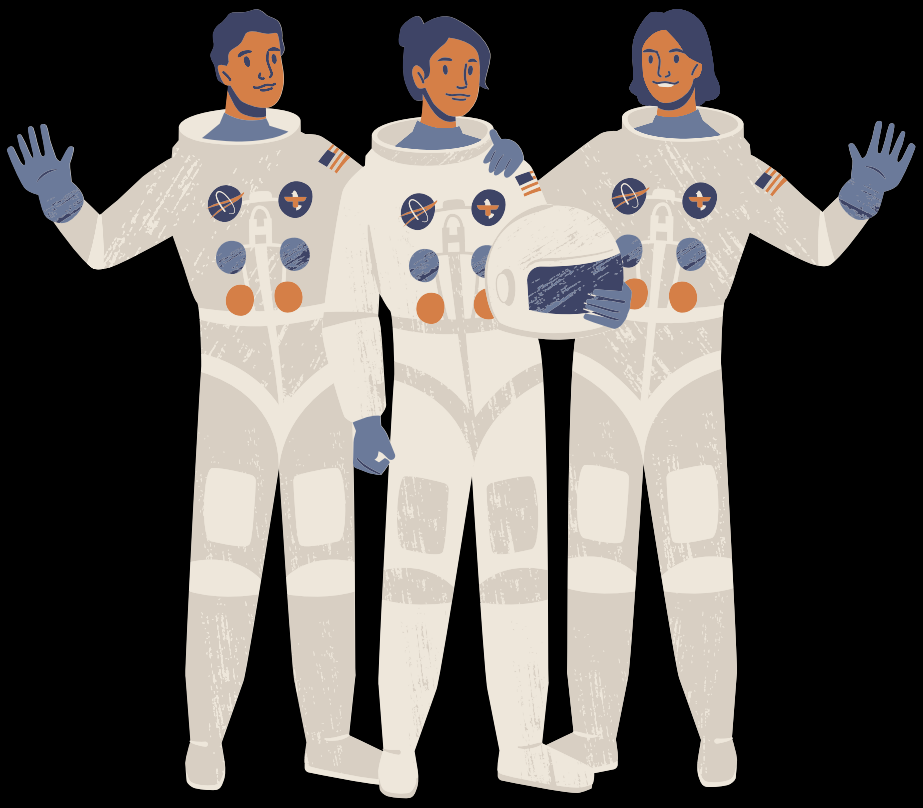
## Have an Idea or Research to Share? Write for our upcoming issue!

Whether you are breaking down the latest astrophysical discoveries, developing machine learning tools for space science, building CubeSat instrumentation, or conducting observational sky surveys—we want your voice in our pages. We invite student researchers, amateur astronomers, and space enthusiasts to submit original articles, project deep-dives, observational logs, and thought-pieces for our upcoming editions.

**Submission categories include:**

- Original observational and computational space research
- Hackathon write-ups, hardware builds, and STEM project showcases
- Opinion pieces on space policy, exploration, and future missions
- Astrophotography features, field notes, and visual guides

Send your drafts, pitches, or inquiries directly to celestaclub@gmail.com. Our editorial team will review your pitch and assist with revisions to help get your work published.



# Space Career

# Astrophysicist

Astrophysicists study the physical nature of stars, galaxies, and the universe at large — how stars are born and die, how galaxies are structured, and the physics of extreme objects like black holes and neutron stars, none of which return calls but all of which leave excellent data behind.

Day to day, the job looks less like gazing wistfully through a telescope and more like a mix of data analysis, coding, statistical modeling, and writing (and rewriting, and rewriting again) research papers — with teaching thrown in for many.

The usual path: a bachelor's degree in physics or astronomy, followed by a PhD for most research roles. Strong programming and math skills matter here at least as much as an interest in the night sky — arguably more, since most modern astrophysics happens in front of a screen, not behind an eyepiece.

Astrophysicists work at universities, national space agencies like ISRO, NASA, and ESA, observatories, and, increasingly, private space and data-analytics companies. And here's the reassuring part: you don't need to become an astronaut to work on space missions. Most people who build, plan, and analyze the data those missions send back never leave the ground — which is, on balance, probably for the best.

# Cosmic Numbers

The universe is roughly 13.8 billion years old, which makes recorded human history look like a rounding error someone forgot to check.

The nearest star  to our own, Proxima Centauri, sits 4.2 light-years away — close, by cosmic standards, and still hopelessly far by any standard a rocket engineer would find comforting.

About 1.3 million Earths could fit inside the Sun, a fact that tends to recalibrate one's sense of personal significance rather quickly.

The International Space Station travels at roughly 17,500 mph (28,000 km/h) to stay in orbit, circling the planet about once every ninety minutes — meaning its crew watches a sunrise roughly every hour and a half and, reportedly, still find it worth watching.

Astronomers have confirmed over 5,800 exoplanets as of 2026, a number that keeps climbing and shows no particular intention of slowing down.

The average temperature of deep space hovers around -455°F (-270°C) — cold enough that "bring a jacket" would be, technically, the understatement of the millennium.

There are an estimated 100 billion or more galaxies in the observable universe, each containing billions of stars of its own — which is either the most humbling statistic in this magazine or an excellent excuse to stop worrying about that email you forgot to send.

And finally: GPS satellites run about 38 microseconds a day faster than clocks on Earth, thanks to time dilation — a strange little debt the universe quietly settles every single day, whether or not anyone's watching.

## About Celesta: Astronomy, Technology, Access

Founded in 2025, Celesta is a student-led organization bridging observational astronomy, computational research, and hands-on STEM education. We exist to make space science deeply accessible without sacrificing its scientific rigor—empowering students to move beyond passive learning into active exploration.

## What We Do

Celesta connects curious minds with real scientific tools. Our initiatives blend theory, software, and engineering to give students practical exposure to modern astrophysics:

- **Real-Data Research**: Analyzing open-source mission data from NASA and global observatories.
- **Competitions & Hackathons:** Hosting targeted technical challenges, from machine learning in astrophysics to satellite design.
- **Workshops & Mentorship:** Running peer-to-peer technical bootcamps, expert sessions, and project incubation.
- **Digital Outreach:** Producing educational content, interactive guides, and visual skywatching breakdowns.

## Connect With Our Community

- **Devpost**: Celesta
- **Discord**: Join our active student research hub
- **Instagram**: @celestaastronomy
- **YouTube**: @CelestaAstronomyClub
- **LinkedIn**: Celesta Club
- **Contact**: celestaclub@gmail.com

